Skills Summary

Triangle Congruence: SSS, SAS, ASA, AAS and HL

Use this reference while completing your worksheet or when studying.

Skill 1 — Name the criterion from the evidence

The five criteria (three pairs each; position decides the name):

- **SSS** — three sides. **SAS** — angle *between* the two sides.
- **ASA** — side *between* the two angles. **AAS** — side *not* between them.
- **HL** — right triangles only: hypotenuses and one leg.

Not criteria: SSA (see Skill 3) and AAA (that is similarity, not congruence).

Example: $\triangle ABC \cong \triangle DEF$ by SSS, with $AB = 4x + 1$ and $DE = 6x - 9$
Step 1. Naming the criterion tells me which parts are congruent, so $\overline{AB}$ and $\overline{DE}$ are corresponding sides and their lengths are equal.
Step 2. $4x + 1 = 6x - 9$ gives $10 = 2x$.
$\Rightarrow$ $\boldsymbol{x = 5}$

Try it: $\triangle PQR \cong \triangle STU$, $PQ = 7x - 2$, $ST = 3x + 22$

check: $\boldsymbol{x = 6}$

Skill 2 — Find the third pair the figure hides

Two triangles in one figure usually share a part, and the shared part is the pair nobody marks.

- **Shared side or angle** — reflexive property of congruence.
- **Crossing segments** — the vertical angles at the crossing are congruent.
- **Parallel sides with a transversal** — alternate interior angles are congruent.

Example: Crossing segments give $m\angle 1 = 2x + 15$ and its vertical angle $4x - 5$
Step 1. The two angles are vertical angles, so the theorem already makes them congruent — I set the measures equal rather than hunting for another given.
Step 2. $2x + 15 = 4x - 5$ gives $20 = 2x$.
$\Rightarrow$ $\boldsymbol{x = 10}$

Try it: Vertical angles measure $5x + 4$ and $3x + 20$

check: $\boldsymbol{x = 8}$

Skill 3 — HL, and why SSA fails without it

SSA is not a criterion: a side of fixed length swung from a fixed angle can reach the far ray in two places, giving two different triangles from the same data.

HL escapes this because a right angle plus the hypotenuse forces the third side: leg $= \sqrt{\text{hyp}^2 - \text{leg}^2}$. Both triangles then have all three sides congruent, so HL is really SSS in disguise.

Example: Right angle at C, hypotenuse 10, leg 6 — find the other leg
Step 1. HL claims two parts determine the whole right triangle, so I check that by computing the third side from just those two.
Step 2. $\sqrt{10^2 - 6^2} = \sqrt{100 - 36} = \sqrt{64}$, and no other value is possible.
$\Rightarrow$ **8**

Try it: Right angle, hypotenuse 25, leg 7 — find the other leg

check: **24**

Skill 4 — Writing the two-column proof

Shape of the proof: list the givens, add the hidden pair (shared part, vertical angles, or alternate interior angles), then close with the criterion and the three statements it uses. Every statement needs a reason; "it looks that way" is not one. CPCTC comes *after* the congruence, never before it.

Example: $ABCD$ has $\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$. Prove $\triangle ABC \cong \triangle CDA$.
Step 1. I have two pairs of sides and the two triangles share the diagonal, so a third side pair is free and SSS is the criterion to aim at.
Step 2. 1. $\overline{AB} \cong \overline{CD}$ (given). 2. $\overline{BC} \cong \overline{DA}$ (given). 3. $\overline{AC} \cong \overline{CA}$ (reflexive). 4. $\triangle ABC \cong \triangle CDA$ by SSS.

Try it: In the same figure, prove $\triangle ABD \cong \triangle CDB$ using the diagonal $\overline{BD}$.

check: same three steps with $\overline{BD} \cong \overline{DB}$ reflexive, then SSS